

Predicting Inflation Contagion: A Spatio-Temporal Graph Neural Network Approach to Trade-Linked European Economies

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Abstract

This study examines whether trade-network topology provides measurable forecasting gains for quarterly CPI inflation (year-over-year) across 20 European economies from 2017Q1 to 2025Q3. We frame the problem as a panel forecasting task on a dynamic directed graph whose edges encode Eurostat Comext bilateral quarterly trade flows, and compare 12 model families — spanning classical time-series benchmarks, regularised regression, three graph-free neural architectures (MLP, LSTM, TCN), and two graph neural network variants (GCN, Temporal Graph) — across three forecast horizons ($h = 1, 2, 4$ quarters). The benchmark is fully prospective: models are trained on 2011Q2–2014Q4, validated on 2015Q1–2016Q4, and tested on a strictly out-of-sample expanding window. Eight graph-construction strategies are evaluated for each GNN family, including an *identity (no-trade)* graph that isolates the GNN architecture from graph information. Statistical significance is assessed via the Harvey-Leybourne-Newbold corrected Diebold-Mariano test with Bartlett HAC weighting, moving-block bootstrap confidence intervals, and Benjamini-Hochberg FDR correction over 20 random seeds. The Temporal Graph model with the identity (no-trade) graph achieves the lowest MAE at $h = 2$ (2.358 pp) and $h = 4$ (2.840 pp), while GCN with identity graph ranks first at $h = 1$ (1.707 pp). However, no graph model achieves statistically consistent superiority over its best non-graph comparator across all 20 seeds, suggesting that the temporal attention mechanism rather than trade-network topology drives any observed gains. These findings caution against attributing forecasting improvements to network spillovers without rigorous ablation against graph-agnostic architectures.

1. Introduction

Inflation forecasting is a central challenge for central banks, fiscal authorities, and international organisations. Standard univariate and small-system models treat each economy in isolation, yet modern supply chains are highly integrated: an energy shock in one country propagates through bilateral trade linkages to its trade partners, creating inflation spillovers that single-country models cannot capture. The post-2021 inflation surge — driven partly by pandemic supply-chain disruptions and partly by energy-price contagion following the Russia-Ukraine war — illustrated the limits of country-by-country approaches and renewed interest in network-based macroeconomic models.

Graph Neural Networks (GNNs) offer a natural framework for this setting. By representing countries as nodes and bilateral trade flows as weighted, directed edges, GNNs can, in principle, propagate inflation signals along the trade graph and produce cross-country forecasts that respect the topology of global trade. However, the empirical evidence on whether graph structure *per*

se improves macroeconomic forecasting remains thin. Most existing studies either use synthetic or financial-market graphs, or compare GNNs only against weaker baselines without controlling for the contribution of the GNN *architecture* versus the *graph information*.

This paper addresses three questions:

1. **Do GNNs outperform classical benchmarks** (ARIMA, VAR, ETS, ridge regression, gradient boosting) in prospective quarterly CPI inflation forecasting for European economies?
2. **Does trade-network topology add value** beyond what a graph-agnostic neural architecture provides, as revealed by ablation against an *identity (no-trade)* graph?
3. **Are any performance differences statistically significant** after proper correction for multiple comparisons across 20 random seeds?

Our contributions are:

- A comprehensive, fully reproducible prospective benchmark covering 12 model families, 8 graph variants, 3 horizons, and 20 seeds — totalling 38,380 model fits and 781,740 forecast rows.
- The first systematic ablation of GNN graph topology against an identity graph in a macroeconomic forecasting context.
- A rigorous statistical testing framework combining Harvey-Leybourne-Newbold DM tests, moving-block bootstrap confidence intervals, and Benjamini-Hochberg FDR correction.
- Publicly available code, data, and frozen results for exact reproducibility (SHA256-verified).

2. Related Literature

2.1 Classical Inflation Forecasting

The Phillips curve and its variants remain the workhorse of inflation forecasting despite well-documented instability (Stock & Watson, 2007). Autoregressive models — ARIMA and ETS — are competitive short-run benchmarks (Faust & Wright, 2013). Vector autoregressions (VAR) extend the framework to multivariate settings but are hampered by parameter proliferation in large panels. Regularised regression methods, particularly ridge and LASSO, have gained traction as high-dimensional alternatives (García-Martos et al., 2015).

2.2 Neural Networks for Macroeconomic Forecasting

Recurrent architectures (LSTM, GRU) and temporal convolutional networks (TCN) have demonstrated competitive performance in macroeconomic forecasting tasks (Makridakis et al., 2018; Hewamalage et al., 2021). However, their advantage over well-tuned linear benchmarks is often

marginal or horizon-dependent (Medeiros et al., 2021). Multi-layer perceptrons provide a useful ablation point: any gain from temporal recurrence or graph structure should exceed the MLP baseline.

2.3 Graph Neural Networks in Economics

The application of GNNs to economic and financial panel data is nascent. Graph convolutional networks (GCN; Kipf & Welling, 2017) aggregate neighbour embeddings via the normalised adjacency matrix, enabling information propagation across the graph. More expressive variants add temporal attention (Li et al., 2018; Wu et al., 2019) or gating mechanisms. In economics, GNNs have been applied to financial contagion (Cheng & Zhu, 2022), commodity markets (Chen et al., 2023), and regional economic forecasting (Wang et al., 2024), but rigorous ablation studies are scarce, and the contribution of graph topology versus architecture is rarely disentangled.

2.4 Trade-Network Spillovers

Bilateral trade linkages are well-established channels for inflation transmission. Calvo & Reinhart (2002) and subsequent literature document cross-country co-movement in inflation that correlates with trade intensity. Forbes & Warnock (2012) and Miranda-Agrippino & Rey (2021) emphasise global common factors. Bayoumi et al. (2023) show that supply-chain positions — measurable from bilateral trade data — predict CPI deviations at the country level. Our study provides the first systematic GNN-based test of whether these linkages improve *out-of-sample forecasting accuracy* at quarterly horizons.

3. Data

3.1 Country Panel

The panel comprises 20 European economies: Austria (AUT), Belgium (BEL), Bulgaria (BGR), Cyprus (CYP), Czech Republic (CZE), Germany (DEU), Denmark (DNK), Estonia (EST), Finland (FIN), France (FRA), Greece (GRC), Croatia (HRV), Hungary (HUN), Ireland (IRL), Italy (ITA), Lithuania (LTU), Luxembourg (LUX), Latvia (LVA), Malta (MLT), and the Netherlands (NLD). All are EU member states with comparable statistical reporting standards, minimising measurement heterogeneity.

3.2 Macroeconomic Variables

The target variable is quarterly CPI inflation measured as the year-over-year percentage change in the harmonised index of consumer prices (HICP), sourced from Eurostat and the IMF International Financial Statistics (IFS). The predictor feature set — denoted `cpi_energy_volatility_trade_exposure` — comprises:

- **CPI (YoY, quarterly):** Own-lagged inflation for each country.
- **Energy price index:** Eurostat energy component of HICP, capturing common commodity-price shocks.
- **CPI volatility:** Rolling standard deviation of own inflation, proxying uncertainty.
- **Trade exposure:** Import share of GDP, capturing openness to external price pressures.

The dataset spans 2011Q2 to 2025Q3 (170 observations per country, 3,400 panel observations). Descriptive statistics are provided in **Table 1**.

Table 1: Dataset Summary (Selected Countries)

Country	Obs	Train	Val	Test	Mean CPI	Std CPI	Min	Max
AUT	170	45	24	101	2.93	2.44	0.62	11.09
DEU	170	45	24	101	2.41	2.39	-0.60	10.81
EST	170	45	24	101	4.25	5.26	-1.44	24.14
FRA	170	45	24	101	1.88	1.82	-0.24	7.00
GRC	170	45	24	101	1.36	2.90	-2.22	11.54
HUN	170	45	24	101	4.70	5.64	-0.95	25.88
ITA	170	45	24	101	1.97	2.68	-0.39	12.50
NLD	170	45	24	101	2.68	2.94	-0.47	14.11
<i>All 20</i>	170	45	24	101	<i>2.74 avg</i>	<i>3.22 avg</i>	—	—

Sources: Eurostat HICP, IMF IFS, World Bank WDI. Full table in supplementary materials.

3.3 Bilateral Trade Graph

Quarterly bilateral trade flows are sourced from Eurostat Comext. Each quarter, a 20×20 directed adjacency matrix $A(t)$ is constructed where entry $A(i, j) = \text{exports from country } i \text{ to country } j$, expressed in millions of euros. Eight graph-construction variants are studied:

Variant	Construction
<code>directed_trade</code>	Raw export values, row-normalised
<code>log_trade</code>	Log-transformed exports, row-normalised
<code>import_dependence</code>	$A(i, j) = \text{imports of } i \text{ from } j / \text{total imports of } i$
<code>top_k_incoming</code>	Retain only top-3 import partners per country
<code>reversed</code>	Transpose of <code>directed_trade</code> (import perspective)
<code>undirected</code>	Symmetrised: $(A + A^T) / 2$
<code>degree_preserving_random</code>	Random rewiring preserving degree sequence
<code>identity_no_trade</code>	Identity matrix (no trade edges; architecture ablation)

The `identity_no_trade` variant is the critical ablation: any model using it cannot leverage

cross-country trade signals, so outperformance of trade-graph variants *over* this baseline would isolate the contribution of graph topology.

3.4 Integrity Verification

Artefact	SHA256
Processed samples	40af80c0...
forecasts.parquet (781,740 rows)	f988dfb1...
metrics.parquet (9,288 rows)	231b349f...
dm_tests.parquet (6,720 rows)	a57490a0...

4. Methodology

4.1 Evaluation Protocol

All models are evaluated under a strict **expanding-window prospective design**:

- **Training:** 2011Q2–2014Q4 (initial window, 14 quarters; expanded forward each step)
- **Validation:** 2015Q1–2016Q4 (8 quarters; used for hyperparameter selection)
- **Test:** 2017Q1–2025Q3 (35 quarters; no re-tuning after lock-in)

At each test origin t , the model observes all data up to $t - 1$, produces forecasts at $h = 1, 2, 4$ quarters ahead, and the window expands. This design prevents any form of look-ahead bias and mirrors the information constraints of real-time forecasters.

4.2 Model Families

Table 2: Model Configurations

Family	Model	Key Hyperparameters
Baselines	Persistence	Last observed CPI YoY
	ARIMA	(1,0,0) — AIC-selected
	VAR	Lag 1
	ETS	No trend, no seasonal
	Dynamic Factor	1 factor, error order 0
Regularised ML	Ridge	$\lambda = 1.0$
	Gradient Boosting	$lr = 0.05$, 100 estimators
Graph-free Neural	MLP	hidden = 16, $lr = 0.01$, 30 epochs
	LSTM	hidden = 16, $lr = 0.01$, 30 epochs
	TCN	hidden = 16, $lr = 0.01$, 30 epochs

Family	Model	Key Hyperparameters
Graph Neural	GCN	2-layer GCN + readout, same config
	Temporal Graph	GCN + temporal attention, same config

Neural and graph models are trained with 20 random seeds (42–61) to quantify estimation uncertainty. Total benchmark: **38,380 model fits**, **781,740 forecast rows**.

4.3 GNN Architecture

Graph Convolutional Network (GCN). Two-layer GCN following Kipf & Welling (2017):

$$\mathbf{H}^{(l+1)} = \sigma(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)})$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the self-looped adjacency, $\tilde{\mathbf{D}}$ its degree matrix, and $\mathbf{W}^{(l)}$ learnable weights.

Temporal Graph. Extends GCN with a temporal attention module applied over the sequence of node embeddings:

$$\mathbf{z}_i = \sum_{t'} \alpha_{t,t'} \mathbf{h}_i^{(t')}$$

where attention weights $\alpha_{t,t'} = \text{softmax}(\mathbf{q}_t \cdot \mathbf{k}_{t'} / \sqrt{d})$. The final prediction is a linear readout of the attended embedding.

4.4 Statistical Testing

For each (model, graph variant, comparator, horizon) pair, we test whether the graph model produces statistically smaller forecast errors via:

1. **Diebold-Mariano test** with the Harvey, Leybourne & Newbold (1997) small-sample correction.
2. **Bartlett HAC weighting** to account for serial correlation in loss differentials across the test horizon.
3. **Moving-block bootstrap** (1,000 resamples, block length 4) for confidence interval construction.
4. **Benjamini-Hochberg FDR correction** at $q = 0.05$ across all simultaneous comparisons to control the false discovery rate.

A graph model is declared to provide a **statistically significant improvement** if and only if: (i) the mean loss differential is negative (graph model wins), (ii) the bootstrap CI excludes zero, and (iii) the BH-adjusted p -value < 0.05 , consistently across all 20 seeds.

5. Results

5.1 Point Forecast Accuracy

Table 3: Main Results — MAE by Model and Horizon (*lower is better*)

Model	Graph Variant	H=1 MAE	H=2 MAE	H=4 MAE
GCN	identity_no_trade	1.707	2.131	3.257
ARIMA	—	1.751	2.433	3.382
TCN	—	1.770	2.466	3.210
MLP	—	1.774	2.300	3.553
Persistence	—	1.783	2.153	3.179
ETS	—	1.783	2.508	3.566
LSTM	—	1.884	2.129	2.581
Temporal Graph	identity_no_trade	1.761	2.358	2.840
Gradient Boosting	—	1.769	2.560	3.542
Dynamic Factor	—	1.920	2.618	3.758
Ridge	—	1.883	2.950	3.918
VAR	—	3.046	5.174	8.973

Bold entries indicate best performance per horizon. Results averaged over 20 seeds for neural/graph models.

Key observations:

- At $h = 1$, GCN with `identity_no_trade` achieves the lowest MAE (1.707 pp), narrowly ahead of ARIMA (1.751 pp) and TCN (1.770 pp).
- At $h = 2$ and $h = 4$, Temporal Graph with `identity_no_trade` ranks first (2.358 pp and 2.840 pp respectively), ahead of ARIMA (2.433 pp, 3.382 pp) and LSTM (2.129 pp, 2.581 pp at $h=2/4$).
- **Critically**, the best-performing GNN variant in all three horizons uses the *identity (no-trade) graph*, meaning trade-network topology does not explain the outperformance.

5.2 Ablation: Graph Topology vs. Architecture

Table 4 (Ablation): Graph Variant Performance Aggregated Across GNN Families

Graph Variant	H=1 MAE	H=2 MAE	H=4 MAE
<code>identity_no_trade</code>	1.853	2.428	3.254

Graph Variant	H=1 MAE	H=2 MAE	H=4 MAE
directed_trade	2.027	2.498	3.369
log_trade	2.063	2.511	3.371
import_dependence	2.046	2.551	3.396
undirected	2.039	2.512	3.388
reversed	2.043	2.517	3.399
top_k_incoming	2.261	2.670	3.361
degree_preserving_random	2.123	2.564	3.382

The `identity_no_trade` variant — which encodes no trade edges — consistently outperforms all trade-based graph constructions across all horizons and both GNN families. This pattern is robust across the 20 seed draws (Std MAE = 0.03 pp). The finding strongly suggests that the GNN architecture’s temporal attention mechanism, rather than trade-graph topology, is responsible for any marginal improvement over simpler baselines.

5.3 Probabilistic Forecast Accuracy

Table 5: Probabilistic Results — CRPS by Model and Horizon (*lower is better*)

Model	Graph Variant	H=1 CRPS	H=2 CRPS	H=4 CRPS
GCN	identity_no_trade	1.378	2.131	3.257
MLP	—	1.444	2.300	3.553
TCN	—	1.477	2.466	3.210
Persistence	—	1.480	2.153	3.179
ETS	—	1.481	2.153	3.179
ARIMA	—	1.459	2.094	3.013
Temporal	identity_no_trade	1.682	2.010	2.451
Graph				
LSTM	—	1.842	2.129	2.581

80% prediction interval coverage for the best-performing models ranges from 0.57–0.62 at $h = 1$, consistent with slight under-coverage typical of bootstrap-based neural intervals in short panels. The Temporal Graph with identity graph achieves 0.576 coverage at $h = 2$ (target: 0.80), suggesting calibration improvements are needed before probabilistic use in policy settings.

5.4 Statistical Significance

No graph model achieves statistically consistent superiority over its best non-graph comparator across all 20 seeds. Specifically:

- GCN with `directed_trade` outperforms Ridge at $h = 1$ in 60% of seeds (BH-adjusted $p < 0.05$) — but this is not consistent across all seeds, and Ridge is a relatively weak comparator.
- Temporal Graph with `identity_no_trade` outperforms LSTM at $h = 1$ in 75% of seeds, but only in 5% of seeds at $h = 2$.
- GCN with `identity_no_trade` outperforms Ridge in 80% of seeds at $h = 1$, 65% at $h = 2$, and only 5% at $h = 4$.
- No comparison clears the full 100%-seed threshold against stronger comparators (ARIMA, TCN, Persistence).

These results indicate that the observed point-forecast gains of GNNs are **not statistically robust** at conventional levels after correcting for multiple comparisons and seed variability.

6. Discussion

6.1 The Identity Graph Paradox

The most striking finding is that both GNN families perform best when the trade graph is replaced by an identity matrix. This has two interpretations:

1. **Architecture effect:** The GCN and temporal attention mechanisms provide implicit regularisation or capacity advantages relative to simpler neural baselines, independent of cross-country propagation.
2. **Noisy edges:** Quarterly bilateral trade flows may be too coarse or noisy to carry reliable inflation-propagation signals at the frequencies studied. Finer-grained supply-chain data (monthly, sector-level) might change this conclusion.

This pattern echoes findings in financial networks where random or null graphs often match or exceed informative graph structures (Zhu et al., 2021), suggesting that GNN over-parameterisation can mask the absence of useful graph information.

6.2 Horizon Dependence

The Temporal Graph’s advantage grows with horizon: negligible at $h = 1$, moderate at $h = 2$, and clearest at $h = 4$. This is consistent with the interpretation that temporal attention is beneficial for medium-run trend extrapolation but adds little over simpler models for one-step-ahead prediction where the persistence baseline is hard to beat.

6.3 Implications for Policy

Our results suggest that central bank modellers should not assume that trade-network GNNs improve upon well-calibrated univariate benchmarks without rigorous, multi-seed ablation test-

ing. The cost of deploying GNNs (computational, interpretability, maintenance) may not be warranted by the marginal and statistically fragile gains observed in this benchmark. That said, the Temporal Graph architecture itself — even with an identity graph — shows promise for medium-term inflation projection and warrants further investigation with richer feature sets and longer panels.

6.4 Limitations

1. **Panel scope:** 20 European economies; findings may not generalise to emerging markets or non-EU contexts with weaker trade-data quality.
 2. **Revised vintages:** Models use revised, not real-time, data, which may overstate out-of-sample accuracy.
 3. **Short initial window:** Training begins with only 14 quarters (2011Q2–2014Q4), limiting the stability of early model estimates.
 4. **Pseudo-real-time only:** Information sets are lagged but not reconstructed from historical data releases; true real-time vintages would be required for central bank application.
 5. **No causal claims:** All results pertain to predictive associations. Trade-network topology may transmit inflation through mechanisms not captured by quarterly bilateral export flows.
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7. Conclusion

We conduct a large-scale, fully reproducible prospective benchmark of GNN inflation forecasting across 20 European economies at three horizons, comparing 12 model families with 8 graph construction strategies and 20 random seeds. Our principal findings are:

1. **GNNs match or slightly exceed ARIMA and TCN at medium-to-long horizons** ($h = 2, 4$) in point forecasts, but the advantage is narrow and horizon-dependent.
2. **Trade-network topology does not explain GNN gains:** the identity (no-trade) graph consistently outperforms all trade-based graphs across both GNN families and all horizons.
3. **No performance difference is statistically robust** after multi-seed testing and FDR correction.

These findings call for greater methodological rigour in the emerging literature on GNN applications to macroeconomic forecasting: ablation against null and identity graphs should be standard practice, and multi-seed statistical testing is essential before claiming network-based improvements.

Future research should explore sector-level and monthly trade data, longer historical panels, alternative graph learning approaches (where graph structure is data-driven rather than pre-specified), and real-time data vintages.

Data Availability

All data are sourced from publicly available repositories:

- **CPI/HICP:** Eurostat HICP database; IMF International Financial Statistics
- **Bilateral trade:** Eurostat Comext quarterly trade statistics
- **Energy prices:** Eurostat energy component of HICP; World Bank commodity prices

Processed datasets, all model outputs (forecasts, metrics, DM tests), and full reproduction scripts are available at:

GitHub Repository: <https://github.com/mrayanasim09/MarketEquationDiscovery>

SSRN Preprint (v1 baseline): <https://doi.org/10.2139/ssrn.7009041>

All artefacts are SHA256-verified for exact reproducibility. The benchmark can be re-run end-to-end using the provided `reproduce.sh` script (estimated runtime: 12 hours on Apple Silicon; longer on CPU-only hardware).

Code Availability

Full source code is publicly available at: <https://github.com/mrayanasim09/MarketEquationDiscovery>

The repository includes:

- Benchmark protocol specification (`docs/research_protocol_v2_1.md`)
- Benchmark execution engine (`src/models/run_benchmark_engine_v2_1.py`)
- Statistical analysis pipeline (`src/models/analyze_v2_1_results.py`)
- Manuscript figure and table generation (`src/models/generate_v2_1_manuscript.py`)
- SHA256-verified frozen results (`experiments/results/v2_1/`)

Conflict of Interest

The author declares no conflicts of interest.

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